

Dynamic Random-Access Memory (DRAM)

■ Key features

- **DRAM** is traditionally packaged as a chip
- Basic storage unit is normally a **cell** (one bit per cell)
- Multiple DRAM chips form main memory in most computers

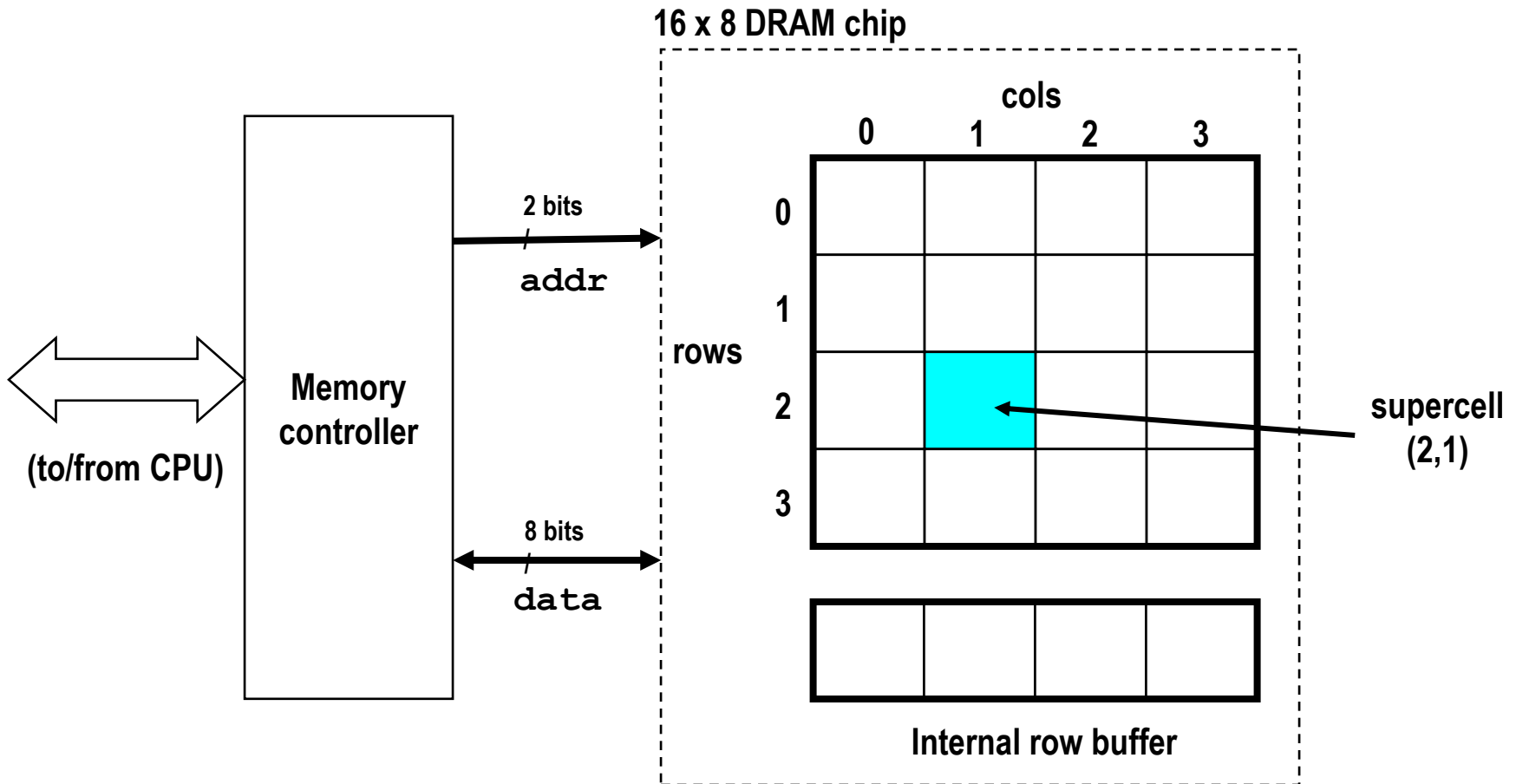
■ Technical characteristics

- Organized in two dimensions (rows and columns)
 - To access (within a DRAM chip): select row then select column
 - Consequence: 2nd access to a row faster than different column/row
- Each cell stores bit with a capacitor; one transistor is used for access
- Value must be refreshed every 10-100 ms
 - Done within the hardware

Conventional DRAM Organization

■ d x w DRAM:

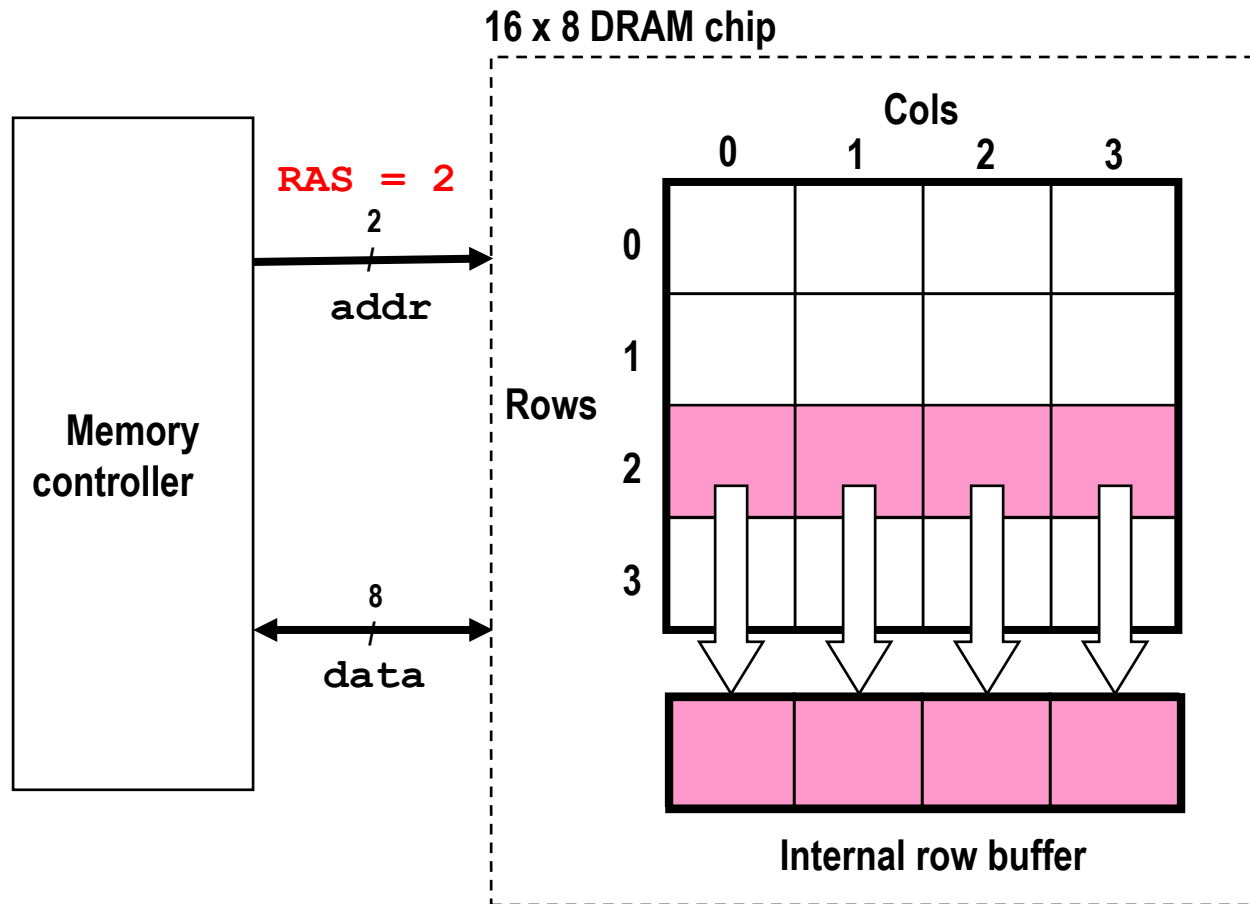
- dw total bits organized as d **supercells** of size w bits



Reading DRAM Supercell (2,1)

Step 1(a): Row access strobe (**RAS**) selects row 2.

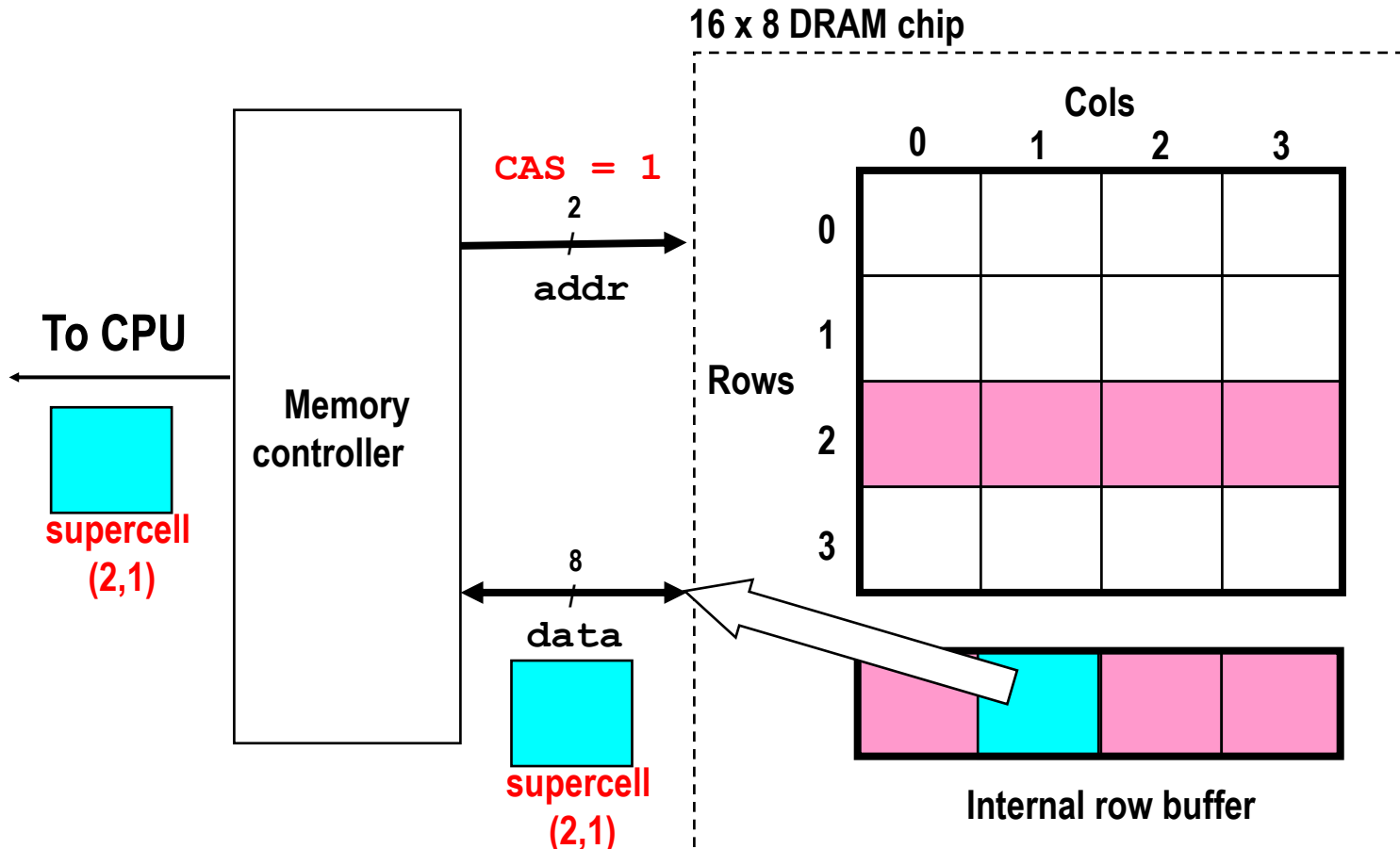
Step 1(b): Row 2 copied from DRAM array to row buffer.



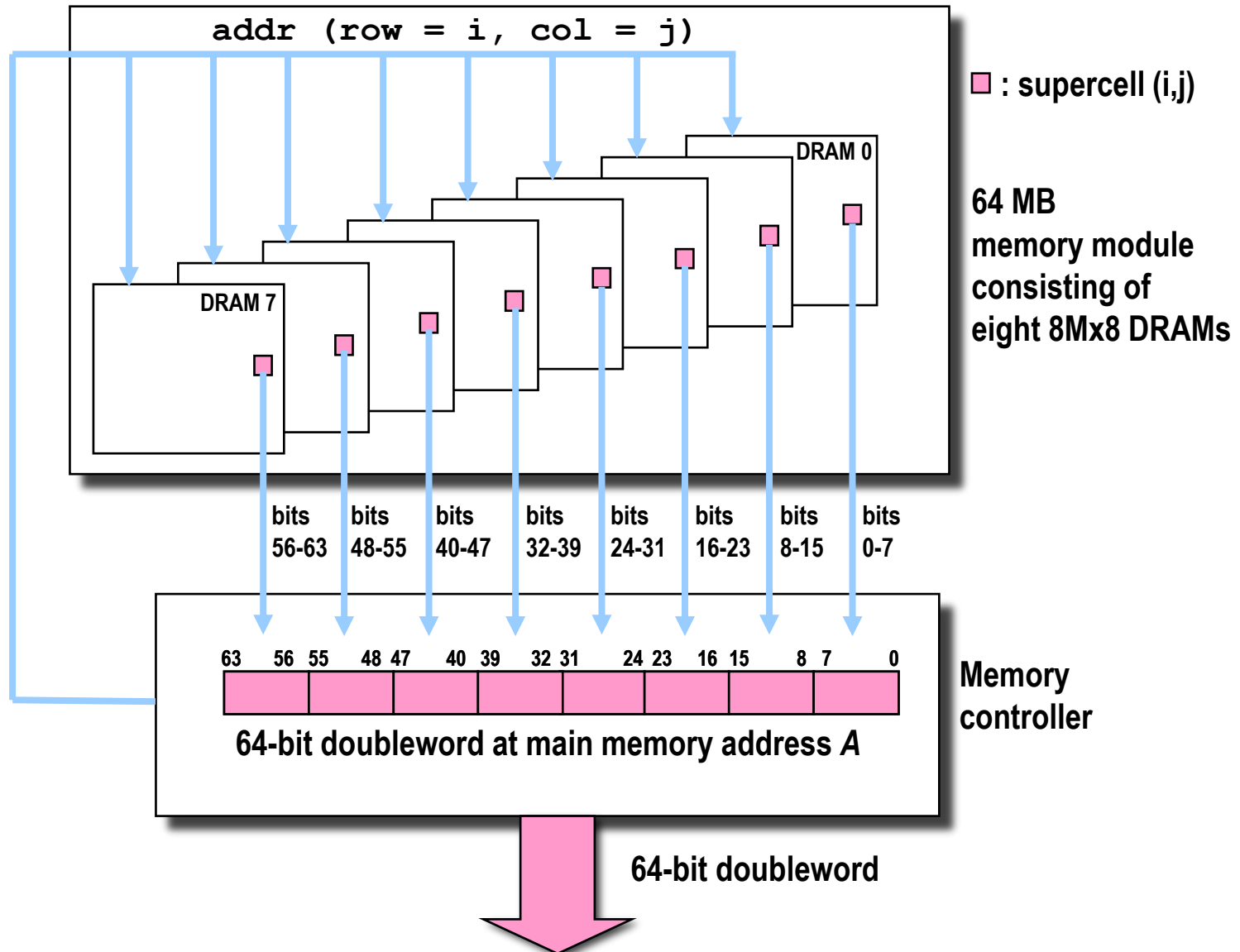
Reading DRAM Supercell (2,1)

Step 2(a): Column access strobe (**CAS**) selects column 1.

Step 2(b): Supercell (2,1) copied from buffer to data lines, and eventually back to the CPU.



Memory Modules



Random-Access Memory (RAM)

■ Key features

- **RAM** is traditionally packaged as a chip.
- Basic storage unit is normally a **cell** (one bit per cell).
- Multiple RAM chips form a memory.

■ Static RAM (SRAM)

- Each cell stores a bit with a four or six-transistor circuit.
- Retains value indefinitely, as long as it is kept powered.
- Relatively insensitive to electrical noise (EMI), radiation, etc.
- Faster and more expensive than DRAM.

■ Dynamic RAM (DRAM)

- Each cell stores bit with a capacitor. One transistor is used for access
- Value must be refreshed every 10-100 ms.
- More sensitive to disturbances (EMI, radiation,...) than SRAM.
- Slower and cheaper than SRAM.

SRAM vs DRAM Summary

	Trans. per bit	Access time	Needs refresh?	Needs EDC?	Cost	Applications
SRAM	4 or 6	1X	No	Maybe	100x	Cache memories
DRAM	1	10X	Yes	Yes	1X	Main memories, frame buffers

Enhanced DRAMs

- **Basic DRAM cell has not changed since its invention in 1966.**
 - Commercialized by Intel in 1970.
- **DRAM cores with better interface logic and faster I/O :**
 - Synchronous DRAM (**SDRAM**)
 - Uses a conventional clock signal instead of asynchronous control
 - Allows reuse of the row addresses (e.g., RAS, CAS, CAS, CAS)
 - Double data-rate synchronous DRAM (**DDR SDRAM**)
 - Double edge clocking sends two bits per cycle per pin
 - Different types distinguished by size of small prefetch buffer:
 - **DDR** (2 bits), **DDR2** (4 bits), **DDR4** (8 bits)
 - By 2010, standard for most server and desktop systems
 - Intel Core i7 supports only DDR3 SDRAM